Cryptocurrency Project Kickoff

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Team Introduction

Gillian Schriever (Master of Data Science in Harvard)

Experience & Focus: Product ownership and public health data systems, with strengths in project management and analytics. Prior consulting experience in accounting company.

Jiahui Cai (Master of Data Science in Harvard)

Experience & Focus: Large language models (LLMs), machine learning, retrieval-augmented systems, causal modeling, statistical experimentation. Focus on product-driven applications

Zhilin Chen (Master of Data Science in Harvard)

Experience & Focus: Large Language Models (LLMs), retrieval-augmented reasoning, and multi-agent chatbot systems. Prior experience in investment banking analytics.

Jinghan Huang (Master of Data Science in Harvard)

Experience & Focus: LLMs and multi-agent AI systems, grounded in Economics and Quantitative Marketing with a focus on data-driven modeling.

Team Strengths: Core strengths in data engineering, statistical modeling, anomaly detection, and financial analytics, supported by structured project management experience

Partner Description

We are collaborating with cryptocurrency and blockchain professionals based in Taiwan (UTC+8).

Industry Expertise

- Ethereum ecosystem and decentralized finance
- Blockchain analytics (Dune Analytics)
- Cryptocurrency investing and entrepreneurship
- Prediction market research and media analysis

Key Industry Contributors

@Lumberg – Entrepreneur and cryptocurrency investor

@MiLLi3e – Ethereum specialist and cofounder of Trueo

@uberjake – Dune Analytics data expert (MIT background)

@eevoy – Harvard alum and published writer on prediction markets (WSJ)

Project Overview

Original Thesis

- Examine time series of cryptocurrency trades/transactions looking for evidence of fraud, market manipulation, abuse, and/or money laundering

Scoped for feasibility in one semester

- **MVP of: Pump-and-dump** detection in **small-cap ERC-20 tokens**
- Focus on abnormal price spikes, volume surges, and ownership concentration patterns.

Why This Problem Matters

- **Small-cap tokens** are particularly vulnerable to manipulation due to **low liquidity**.
- **Pump-and-dump schemes** undermine market transparency and investor trust.
- **Identifying early signals** can contribute to more data-driven crypto market analysis.

Data & Analytical Resources

Data Sources

On-Chain Data (Ethereum / Dune Analytics)

- ERC-20 token transfer history
- Wallet-level transaction activity
- Token ownership concentration metrics

Analytical Approach

- Time-series analysis of price & volume dynamics
- Detection of abnormal spikes and volatility shifts
- Anomaly detection for unusual trading behavior
- Network analysis of wallet interaction patterns

Additional Support

(As Needed)

- Industry guidance on selecting representative small-cap ERC-20 tokens
- Feedback on defining manipulation indicators
- Supplementary datasets via industry collaborators (if required)

Project Deliverables & Timeline

Phase 01

Weeks 1–2

Scope & alignment

Finalize scope, confirm token universe and data access, and align with Lumberg & team on operational definition of pump-and-dump and project success criteria

Phase 02

Weeks 3–4

Data pipeline & exploration

Build reproducible data pipelines and clean OHLCV + on-chain ERC-20 datasets, delivering an initial exploratory data analysis for validation.

Phase 03

Weeks 5–6

Pump episode detection

Develop and validate a price/volume anomaly detector to generate and visualize Top-N suspicious pump episodes.

Phase 04

Weeks 7–8

Structural wallet analysis

Implement on-chain exchange inflow and concentration metrics to assess pre-pump wallet behavior and structural accumulation patterns.

Project Deliverables & Timeline

MONTH YEAR

Phase 05

Weeks 9-10

Unified alert scoring

Integrate market and on-chain signals into a unified alert score and produce 2–3 detailed case studies with uncertainty assessment.

Phase 06

Weeks 11-12

MVP surveillance dashboard

Deliver a lightweight MVP surveillance dashboard/report with interpretable alert outputs and documented limitations.

Phase 07

Weeks 13–14

Reproducible research package

Present final deck, technical appendix, reproducible codebase, and ranked suspicious episode dataset with roadmap for future scaling.

Phase 08

Project Wrap-Up

Validation and synthesis

Conduct final validation, document limitations and uncertainty assessment, archive reproducible materials, and summarize key insights with proposed directions for future research.

Thank You